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Sequential Bayesian Inference for Stochastic Epidemic Models of Cumulative Incidence

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Abstract. Epidemics are inherently stochastic, and stochastic models provide an appropriate way to describe and analyse such phenomena. Given temporal incidence data consisting of, for example, the number of new infections or removals in a given time window, a continuous-time discrete-valued Markov process provides a natural description of the dynamics of each model component, typically taken to be the number of susceptible, exposed, infected or removed individuals. Fitting the SEIR model to time-course data is a challenging problem due incomplete observations and, consequently, the intractability of the observed-data likelihood. Whilst sampling based inference schemes such as Markov chain Monte Carlo are routinely applied, their computational cost typically restricts analysis to data sets of no more than a few thousand infective cases. Instead, we develop a sequential inference scheme that makes use of a computationally cheap approximation of the most natural Markov process model. Crucially, the resulting model allows a tractable conditional parameter posterior which can be summarised in terms of a set of low dimensional statistics. This is used to rejuvenate parameter samples in conjunction with a novel bridge construct for propagating state trajectories conditional on the next incidence count. The resulting inference framework also allows for stochastic infection and reporting rates. We illustrate our approach using synthetic and real data applications.

Keywords: particle filter, S(E)IR model, Markov jump process, time discretisation, bridge construct.

1 Introduction

Statistical epidemiology is a large and ever-growing field that combines statistical models with state-of-the-art inference techniques to analyse complex data arising from observational studies. Epidemics are inherently random, and capturing this aspect necessitates the use of stochastic models, which typically take the form of a continuous-time, discrete-valued Markov jump process (MJP), describing transitions between different states such as susceptible, exposed, infected, removed (SEIR, see e.g. Allen, 2017). Inference for the parameters governing such models is complicated by the availability of incomplete and imperfect observation regimes, as arising, for example, when only a subset of states or

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transitions are recorded at discrete times. This problem can be circumvented via the use of Markov chain Monte Carlo and data augmentation (see e.g. O'Neill and Roberts, 1999; Jewell et al., 2009) although these schemes appear limited to applications where the total population size is of the order of a few thousand individuals (Stockdale et al., 2021). Therefore, a key challenge is the development of fast and reliable inference techniques to allow real-time decision making (Swallow et al., 2022).

Computationally efficient approaches to inference for stochastic epidemic models include the use of approximate inference schemes (see e.g. Kypraios et al., 2017; McKinley et al., 2018; Minter and Retkute, 2019, for approximate Bayesian computation schemes) or direct approximation of the most natural MJP (see e.g. Cauchemez and Ferguson, 2008; Fuchs, 2013; Fintzi et al., 2022, for approximations based on stochastic differential equations) and direct approximation of the observed-data likelihood (see e.g. Whiteley and Rimella, 2021; Golightly et al., 2023). In this paper, we replace the MJP model of cumulative incidence with a model in which the number of transition events over a time interval whose length is chosen by the practitioner is assumed to be Poisson distributed. The resulting time-discretised SEIR model (dSEIR) has a number of advantages over other approximate models, including recognition of the discrete stochastic nature of epidemic spread, easy incorporation of time-varying parameters, specification of a time step that trades off accuracy and computational efficiency and, crucially, tractability of the conditional posterior for rate parameters, given a particular choice of prior.

We fit the dSEIR model within a sequential Bayesian framework. That is, we update our inferences about the static parameters and latent dynamic process as each observation becomes available. The observations are noisy measurements of incidence, specifically, the number of new infections or removals in fixed-length time windows. The inference procedure uses sequential Monte Carlo (see e.g. Kantas et al., 2009; Chopin et al., 2020) via recursive application of a series of propagate, weight, resample and rejuvenate steps with the latter used to lessen sample impoverishment of the static parameter particle set. One such approach, SMC^2 (Chopin et al., 2013), weights parameter particles by an estimate of observed data likelihood, computed via a particle filter over states, and rejuvenates parameter samples through the use of a particle marginal Metropolis-Hastings (PMMH) kernel (see e.g. Andrieu et al., 2010). However, the use of nested particle filters in this way can be computationally expensive.

In this work, we aim to lessen sample impoverishment by leveraging the tractability of the parameter posterior conditional on the latent dynamic process, summarised through a collection of summary statistics. This approach was first described in Storvik (2002) (see also Fearnhead, 2002) and is a key ingredient of particle learning (Carvalho et al., 2010), which has been used by Dukic et al. (2012) and Lin and Ludkovski (2014), among others, in the context of stochastic epidemic models. Our approach is most related to the latter, which considers a particle learning algorithm for the most natural MJP representation of an epidemic compartment model of prevalence data, coupled with a latent seasonal component. However, unlike their approach, our modelling framework allows for additional flexibility via the use of a stochastic differential equation (SDE) to describe time-varying parameters (e.g. infection and/or reporting rates) and the specification of an observation model to allow for imperfect observations of incidence. Moreover, we adapt the bridge construct of Golightly and Wilkinson (2015) (see

also Golightly and Kypraios, 2018) to the context of incidence counts, so that particle trajectories are propagated conditional on the following observation. We illustrate the resulting inference scheme in three scenarios: the first uses synthetic data in order to assess the accuracy of our approach for different numbers of particles (bench-marked against the output of a PMMH scheme) while the two final scenarios consider the spread of Ebola in West Africa (Fintzi et al., 2022) and COVID-19 in New York (Spannaus et al., 2022).

The remainder of this paper is organised as follows. Section 2 describes the Markov jump process representation of cumulative incidence in an SEIR setting before considering the time-discretised approximation. Section 3 outlines our sequential approach to Bayesian inference and gives the bridge construct. Applications are considered in Section 4, and conclusions are drawn in Section 5.

2 Stochastic epidemic models

We consider susceptible, infected, removed (SIR, Andersson and Britton, 2012; Kermack and McKendrick, 1927) and SEIR models (Hethcote, 2000) within which a population of fixed size N_{pop} is classified into compartments consisting of susceptible (S), exposed (E), infectious (I) and removed (R) individuals. In what follows, we describe the stochastic SEIR model as the most general case, and make clear how the SIR model can be obtained as a simplification thereof.

Transitions between compartments can be described by the set of pseudo-reactions given by

$$S + I \xrightarrow{\beta} E + I, \quad E \xrightarrow{\kappa} I, \quad I \xrightarrow{\gamma} R.$$

The first transition describes contact of an infective individual with a susceptible and with the net effect resulting in an exposed individual and one fewer susceptible. The second transition moves an exposed individual to the infected class, and the final transition accounts for an infected individual's removal (recovered with immunity, quarantined or dead). The components of $\theta = (\beta, \kappa, \gamma)'$ denote contact, infection and removal rates. Note that setting $\kappa = 0$ and substituting E for I gives the set of pseudo-reactions governing the SIR model as

$$S + I \xrightarrow{\beta} 2I, \quad I \xrightarrow{\gamma} R,$$

for which β now denotes infection rate.

Let $X_t = (S_t, E_t, I_t)'$ denote the numbers in each state at time $t \geq 0$ and note that $R_t = N_{\text{pop}} - S_t - E_t - I_t$ for all $t \geq 0$. The dynamics of $\{X_t, t \geq 0\}$ are most naturally described by a Markov jump process (MJP), that is, a continuous time, discrete valued Markov process. Assuming that at most one event can occur over a time interval $(t, t + \Delta t]$ and that the state of the system at time t is $x_t = (s_t, e_t, i_t)'$, the MJP is characterised by transition probabilities of the form

$$\mathbb{P}(X_{t+\Delta t} = (s_t - 1, e_t + 1, i_t)' | x_t, \theta) = \beta s_t i_t \, \Delta t + o(\Delta t),$$

$$\begin{aligned}
\mathbb{P}(X_{t+\Delta t} = (s_t, e_t - 1, i_t + 1)' | x_t, \theta) &= \kappa e_t \, \Delta t + o(\Delta t), \\
\mathbb{P}(X_{t+\Delta t} = (s_t, e_t, i_t - 1)' | x_t, \theta) &= \gamma i_t \, \Delta t + o(\Delta t), \\
\mathbb{P}(X_{t+\Delta t} = (s_t, e_t, i_t)' | x_t, \theta) &= 1 - (\beta s_t i_t + \kappa e_t + \gamma i_t) \, \Delta t + o(\Delta t),
\end{aligned}$$

and $o(\Delta t)/\Delta t \to 0$ as $\Delta t \to 0$. Similarly, the cumulative incidence of contact, infection and removal events $\{N_t, t \geq 0\}$ is an MJP governed by the transition probabilities

$$\begin{aligned}
\mathbb{P}(N_{t+\Delta t} = (n_{se} + 1, n_{ei}, n_{ir})' | n_t, x_t, \theta) &= \beta s_t i_t \, \Delta t + o(\Delta t), \\
\mathbb{P}(N_{t+\Delta t} = (n_{se}, n_{ei} + 1, n_{ir})' | n_t, x_t, \theta) &= \kappa e_t \, \Delta t + o(\Delta t), \\
\mathbb{P}(N_{t+\Delta t} = (n_{se}, n_{ei}, n_{ir} + 1)' | n_t, x_t, \theta) &= \gamma i_t \, \Delta t + o(\Delta t), \\
\mathbb{P}(N_{t+\Delta t} = (n_{se}, n_{ei}, n_{ir})' | n_t, x_t, \theta) &= 1 - (\beta s_t i_t + \kappa e_t + \gamma i_t) \, \Delta t + o(\Delta t),
\end{aligned}$$

where $n_t = (n_{se}, n_{ei}, n_{ir})'$ denotes the contact, infection and removal events.

It will be helpful to define the instantaneous rate or hazard function $h(x_t) = (h_1(x_t), h_2(x_t), h_3(x_t))'$ by

$$\begin{aligned}
h(x_t) &= \lim_{\Delta t \to 0} \mathbb{P}(X_{t+\Delta t} | x_t, \theta)/\Delta t \\
&= (\beta s_t i_t, \kappa e_t, \gamma i_t)'
\end{aligned}$$

and we suppress the dependence of the hazard function on θ to simplify the notation. Additionally, we define the net effect matrix by

$$A = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{pmatrix}$$

where it is understood that each row describes the effect on each component of $X_t = (S_t, E_t, I_t)'$ by the respective occurrence of a contact, infection or removal event. Then, the processes describing the prevalence $\{X_t, t \geq 0\}$ and cumulative incidence $\{N_t, t \geq 0\}$ are linked via the equation

$$X_t = x_0 + \sum_i A_i' N_{i,t}$$

where x_0 represents the initial number of susceptible, exposed and infected individuals, A_i denotes the ith row of A and $N_{i,t}$ denotes the ith component of the incidence process at time t.

Given x_0 and θ, generating exact realisations of the incidence process, and therefore the stochastic SEIR model, is straightforward and can be achieved by using well-known simulation algorithms from the stochastic kinetic model literature (see, e.g. Wilkinson, 2018). The simplest approach is Gillespie's direct method (Gillespie, 1977), which simulates the time to the next event as an exponential random variable with rate $\sum_{i=1}^{3} h_i$. The event that occurs will be of type i (with 1 = contact, 2 = infection, 3 = removal) with probability proportional to h_i.

2.1 Time-varying contact rate

In practice, assuming that the contact rate in the SEIR model (or infection rate in the SIR model) remains constant throughout the epidemic may be unreasonable. We therefore follow Dureau et al. (2013); Spannaus et al. (2022); Wadkin et al. (2022) among others and describe the contact rate via an Itô stochastic differential equation (SDE). Let $\{\beta_t, t \geq 0\}$ denote the contact process and consider $\tilde{\beta}_t = \log(\beta_t)$, assumed to satisfy a time-homogeneous SDE, parameterised by λ and of the form

$$\mathrm{d}\tilde{\beta}_t = a(\tilde{\beta}_t, \lambda)\mathrm{d}t + b(\tilde{\beta}_t, \lambda)\mathrm{d}W_t \tag{1}$$

where $\{W_t, t \geq 0\}$ is a standard Brownian motion process. Choice of the drift and diffusion functions $a(\cdot)$ and $b(\cdot)$ in Equation (1) determine the properties of $\tilde{\beta}_t$. For example, $a(\tilde{\beta}_t, \lambda) = 0$ and $b(\tilde{\beta}_t, \lambda) = \lambda$ gives a generalised Brownian motion process (as used in Sections 4.1 and 4.3) which admits an analytic solution over a time interval $(t, t + \Delta t]$ as

$$\tilde{\beta}_{t+\Delta t} = \tilde{\beta}_t + \lambda^{-1/2}\Delta W_t \tag{2}$$

where $\Delta W_t := (W_{t+\Delta t} - W_t) \sim \mathrm{N}(0, \Delta t)$ and λ is a precision parameter. In cases where the SDE (1) cannot be solved analytically, a numerical approximation can be sought. The simplest such approximation is given by the Euler-Maruyama discretisation

$$\tilde{\beta}_{t+\Delta t} = \tilde{\beta}_t + a(\tilde{\beta}_t, \lambda)\Delta t + b(\tilde{\beta}_t, \lambda)\Delta W_t. \tag{3}$$

Replacing β with a time-varying $\beta_t = \exp(\tilde{\beta}_t)$ dictates that the hazard h_1 of the contact reaction is no longer constant between event occurrences. Generating exact realisations from the resulting SEIR model is no longer straightforward unless β_t can be bounded above, permitting the use of Poisson thinning (Lewis and Shedler, 1979). In what follows, we consider a time-discretised stochastic SEIR model, and subsequently base our inferential approach on the resulting approximation. In particular, the practitioner can choose the discretisation level to balance accuracy and computational efficiency.

2.2 Time-discretised model

Consider a time interval $(t, t + \Delta t]$ and denote by $\Delta N_t = (\Delta N_{1,t}, \Delta N_{2,t}, \Delta N_{3,t})'$ the length-3 vector containing the number of events of each type (contact, infection, removal) over this interval. We assume that Δt is small enough to reasonably assume that $\tilde{\beta}_s \approx \tilde{\beta}_t$ and $X_s \approx x_t$ for $s \in (t, t + \Delta t)$. It should then be clear that the ith component of ΔN_t follows a Poisson distribution with rate $h_i(x_t, \tilde{\beta}_t)\Delta t$, where we now let the hazard function depend additionally on $\tilde{\beta}_t$. Explicitly,

$$\Delta N_{1,t} \sim \mathrm{Po}(\exp(\tilde{\beta}_t)\, s_t i_t \Delta t), \quad \Delta N_{2,t} \sim \mathrm{Po}(\kappa\, e_t \Delta t), \quad \Delta N_{3,t} \sim \mathrm{Po}(\gamma\, i_t \Delta t).$$

Hence, we obtain the time-discretised SEIR model (dSEIR) with time-varying contact rates as

$$\begin{cases} X_{t+\Delta t} = x_t + A'\Delta N_t, \\ \tilde{\beta}_{t+\Delta t} = \tilde{\beta}_t + a(\tilde{\beta}_t, \lambda)\Delta t + b(\tilde{\beta}_t, \lambda)\Delta W_t. \end{cases} \tag{4}$$

Simulating exact realisations from dSEIR is now straightforward; see Algorithm 1 for further details.

Algorithm 1 dSEIR: Forward Simulation.

Input: parameters $\theta = (\kappa, \gamma)'$, contact rate process parameters λ, initial conditions $x_0 = (s_0, e_0, i_0)'$ and $\tilde{\beta}_0$, time step Δt and end time $T = m\Delta t$.
For $j = 0, \ldots, m-1$:

1. Set $t := j\Delta t$ and calculate the hazard function $h(x_t, \tilde{\beta}_t) = (\exp(\tilde{\beta}_t)\, s_t i_t, \kappa\, e_t, \gamma\, i_t)'$.
2. Simulate the incidence increment ΔN_t by drawing $\Delta N_{i,t} \sim \text{Po}(h_i \Delta t)$, $i = 1, 2, 3$. Set $x_{t+\Delta t} = x_t + A'\Delta N_t$.
3. Simulate $\Delta W_t \sim \text{N}(0, \Delta t)$ and set $\tilde{\beta}_{t+\Delta t} = \tilde{\beta}_t + a(\tilde{\beta}_t, \lambda)\Delta t + b(\tilde{\beta}_t, \lambda)\Delta W_t$

Output: trajectory $\{(x_t, \tilde{\beta}_t), t = 0, \Delta t, \ldots, T\}$.

3 Sequential Bayesian inference

In this section, we consider the problem of performing full Bayesian inference for the parameters (and unobserved dynamic processes) governing the dSEIR model based on incidence observations that may be incomplete and subject to measurement error. We describe the observation model before considering the inference task. We perform sequential Bayesian inference via a particle filter, a key ingredient of which is a novel construct that allows approximate draws of the dSEIR process between observation instants.

3.1 Observation model

Consider data $y = (y_{t_1}, \ldots, y_{t_L})'$ on a regular time grid, where y_{t_i} is a (partial) observation on the incidence $\Delta N_{t_i} = N_{t_i} - N_{t_{i-1}}$ over a time interval $(t_{i-1}, t_i]$. We assume that the incidence process is not observed directly and consider two observation models as follows. The first model reflects the assumption that cases are under-reported, leading to

$$Y_{t_i} | \Delta N_{t_i} \sim \text{Bin}(P'\Delta N_{t_i}, \rho), \quad i = 1, \ldots, L. \tag{5}$$

Here, P is a constant matrix allowing for observation of a subset of components of ΔN_{t_i} and ρ controls the accuracy of the observation process. In practice, we take $P' = (0, 1, 0)$ in the SEIR model and $P' = (1, 0)$ in the case of the SIR model so that observations are noisy counts of new infections in a given time window. The second observation model allows for over-dispersion in the observation process via

$$Y_{t_i} | \Delta N_{t_i} \sim \text{NegBin}(\mu_i = \rho P' \Delta N_{t_i}, \sigma_i^2 = \mu_i + \mu_i^2/\nu), \quad i = 1, \ldots, L \tag{6}$$

where $\text{NegBin}(\mu, \sigma^2)$ denotes a Negative Binomial random variable with mean μ and variance σ^2. For simplicity, we have assumed that both observation models use a constant reporting rate ρ. We relax this assumption in Section 4.3 by considering a dynamic reporting rate, assumed to follow a (discretised) SDE; see Section 4.3 for further details.

We denote by ϕ the parameters governing the observation model and explicitly link the data to the latent incidence process via the mass function $\pi(y_{t_i} | \Delta n_{t_i}, \phi)$.

3.2 Inference task

We assume that interest lies in the vector of all static parameters $\psi = (\theta', \lambda', \phi')'$, the latent incidence process $\{N_t, t_0 \leq t \leq t_L\}$, contact rate process $\{\tilde{\beta}_t, t_0 \leq t \leq t_L\}$ and the initial state vector x_{t_0}. Note that the initial state vector and incidence process are sufficient to determine the prevalence process $\{X_t, t_0 \leq t \leq t_L\}$ deterministically, through recursive application of the first equation in (4). In what follows, we drop explicit dependence of the incidence process on x_{t_0} from the notation for ease of exposition.

Recall that y_{t_i} denotes a noisy and incomplete observation on the incidence ΔN_{t_i} over $(t_{i-1}, t_i]$. We assume that the inter-observation intervals $\Delta t = t_i - t_{i-1}$ are such that the event hazards over these intervals cannot be assumed plausibly constant. We, therefore partition each time interval as

$$t_{i-1} = \tau_{i,0} < \tau_{i,1} < \cdots < \tau_{i,m} = t_i$$

with $\tau_{i,j} - \tau_{i,j-1} = \Delta\tau = \Delta t/m$. This allows the practitioner to choose the discretisation level m to balance the accuracy and computational efficiency of the dSEIR model (4).

We let $\tilde{\beta} = (\tilde{\beta}_{\tau_{1,0}}, \tilde{\beta}_{\tau_{1,1}}, \ldots, \tilde{\beta}_{\tau_{L,m}})'$ and $\Delta n = (\Delta n_{\tau_{1,1}}, \Delta n_{\tau_{1,2}}, \ldots, \Delta n_{\tau_{L,m}})'$ denote the latent contact rate process and collection of incidences over sub-intervals $(\tau_{i,j-1}, \tau_{i,j}]$, for $i = 1, \ldots, L$ and $j = 1, \ldots, m$. Note that $\Delta n_{t_i} = \sum_{j=1}^{m} \Delta n_{\tau_{i,j}}$ then gives the cumulative incidence over $(t_{i-1}, t_i]$. Upon ascribing a prior density $\pi(\psi)$ to ψ, the inference tasks proceeds via the joint posterior

$$\pi(\psi, \tilde{\beta}, \Delta n|y) \propto \pi(\psi)\pi(\tilde{\beta}|\lambda)\pi(\Delta n|\tilde{\beta}, \theta)\pi(y|\Delta n, \phi). \tag{7}$$

Here, the joint density of the latent contact rate process is

$$\begin{aligned}\pi(\tilde{\beta}|\lambda) &= \pi(\tilde{\beta}_{\tau_{1,0}}) \prod_{i=1}^{L}\prod_{j=1}^{m} \pi(\tilde{\beta}_{\tau_{i,j}}|\tilde{\beta}_{\tau_{i,j-1}}, \lambda) \\ &= \pi(\tilde{\beta}_{\tau_{1,0}}) \prod_{i=1}^{L}\prod_{j=1}^{m} \mathrm{N}(\tilde{\beta}_{\tau_{i,j}}; a(\tilde{\beta}_{\tau_{i,j-1}}, \lambda)\Delta\tau\,,\, b^2(\tilde{\beta}_{\tau_{i,j-1}}, \lambda)\Delta\tau),\end{aligned}$$

where $\mathrm{N}(\cdot; m, v^2)$ denotes the density of a normal random variable with mean m and variance v^2. The joint probability of the incidence process is

$$\pi(\Delta n|\tilde{\beta}, \theta) = \prod_{i=1}^{L}\prod_{j=1}^{m}\prod_{k=1}^{3} \mathrm{Po}(\Delta n_{k,\tau_{i,j}}; h_k(x_{\tau_{i,j-1}}, \tilde{\beta}_{\tau_{i,j-1}})\Delta\tau)$$

where $\mathrm{Po}(\cdot; h)$ denotes the probability mass function of a Poisson random variable with mean h. Finally,

$$\pi(y|\Delta n, \phi) = \prod_{i=1}^{L} \pi(y_{t_i}|\Delta n_{t_i}, \phi)$$

with $\pi(y_{t_i}|\Delta n_{t_i}, \phi)$ as the probability mass function arising from either (5) or (6).

Since the joint posterior in (7) will be intractable, we resort to Monte Carlo methods for generating samples of the parameters and latent dynamic processes. In particular, we wish to perform inference sequentially and develop a particle filter approach in the next section.

3.3 Particle filter approach

It will be helpful here to introduce the shorthand notation $y_{[1,i]} = (y_{t_1}, \ldots, y_{t_i})'$ to denote the observations up to (and including) time t_i. Similarly, $\tilde{\beta}_{[0,i]} = (\tilde{\beta}_{\tau_{1,0}}, \tilde{\beta}_{\tau_{1,1}}, \ldots, \tilde{\beta}_{\tau_{i,m}})'$ and $\Delta n_{(0,i]} = (\Delta n_{\tau_{1,1}}, \Delta n_{\tau_{1,2}}, \ldots, \Delta n_{\tau_{i,m}})'$ denote respectively, the latent contact rate process and collection of incidence counts over the corresponding time horizon.

In addition to the dynamic processes described above, we also require sequential inference of the parameter vector $\psi = (\theta', \lambda', \phi')'$ and recall that $\theta = (\kappa, \gamma)'$ parameterises the SEIR transmission process (infection and removal rate), λ parameterises the time-varying contact rate and ϕ governs the observation model. Now, by applying Bayes theorem sequentially, we obtain the posterior distribution at time t_{i+1} as

$$\begin{aligned}\pi(\psi, \tilde{\beta}_{[0,i+1]}, \Delta n_{(0,i+1]} | y_{[1,i+1]}) \propto \pi(\psi, \tilde{\beta}_{[0,i]}, \Delta n_{(0,i]} | y_{[1,i]}) \pi(\tilde{\beta}_{(i,i+1]} | \tilde{\beta}_{t_i}, \lambda) \\ \times \pi(\Delta n_{(i,i+1]} | \tilde{\beta}_{[i,i+1)}, \theta) \pi(y_{t_{i+1}} | \Delta n_{t_{i+1}}, \phi) \qquad (8)\end{aligned}$$

and note that $\tilde{\beta}_{(i,i+1]} = (\tilde{\beta}_{\tau_{i+1,1}}, \ldots, \tilde{\beta}_{\tau_{i+1,m}})'$, $\tilde{\beta}_{[i,i+1)} = (\tilde{\beta}_{\tau_{i+1,0}}, \ldots, \tilde{\beta}_{\tau_{i+1,m-1}})'$ and $\tilde{\beta}_{t_i} = \tilde{\beta}_{\tau_{i,m}} = \tilde{\beta}_{\tau_{i+1,0}}$. Given an equally weighted sample of 'particles' of size N, $\left(\psi^{(1:N)}, \tilde{\beta}_{[0,i]}^{(1:N)}, \Delta n_{(0,i]}^{(1:N)}\right)$, from $\pi(\psi, \tilde{\beta}_{[0,i]}, \Delta n_{(0,i]} | y_{[1,i]})$, the form of (8) immediately suggests an importance resampling step that extends the dynamic process particles by propagating $\tilde{\beta}_{(i,i+1]}^{(k)} \sim \pi(\cdot | \tilde{\beta}_{t_i}^{(k)}, \lambda^{(k)})$ and $\Delta n_{(i,i+1]}^{(k)} \sim \pi(\cdot | \tilde{\beta}_{[i,i+1)}^{(k)}, \theta^{(k)})$, then weights the resulting particles $\left(\psi^{(k)}, \tilde{\beta}_{[0,i+1]}^{(k)}, \Delta n_{(0,i+1]}^{(k)}\right)$ by $w_{i+1}^{(k)} \propto \pi(y_{t_{i+1}} | \Delta n_{t_{i+1}}^{(k)}, \phi^{(k)})$, and finally samples with replacement among the particles (using the weights as probabilities). In what follows, we assume that interest lies in the sequence of filtering distributions $\pi(\psi, \tilde{\beta}_{t_i}, \Delta n_{t_i} | y_{[1,i]})$, $i = 1, \ldots, L$, so that only particles $\left(\psi^{(1:N)}, \tilde{\beta}_{t_i}^{(1:N)}, \Delta n_{t_i}^{(1:N)}\right)$ need be stored at each observation time. Applying the propagate, weight and resample steps recursively in time, gives the bootstrap particle filter (see e.g. Gordon et al., 1993). Two problems are apparent here. Firstly, repeated resampling of static parameter particles will lead to sample impoverishment (with marginal parameter posteriors collapsing to point masses). Secondly, drawing the incidence process 'blindly' (that is, via forward simulation) is likely to lead to many particle trajectories with negligible weight.

To lessen the problem of sample impoverishment, we follow the approach of Storvik (2002) (see also Fearnhead (2002)) by noting that the conditional posterior of a subset of parameter components is tractable for a particular choice of prior. To this end, consider the factorisation

$$\begin{aligned}\pi(\psi, \tilde{\beta}_{[0,i]}, \Delta n_{(0,i]} | y_{[1,i]}) &= \pi(\theta, \lambda | \phi, \tilde{\beta}_{[0,i]}, \Delta n_{(0,i]}, y_{[1,i]}) \pi(\phi, \tilde{\beta}_{[0,i]}, \Delta n_{(0,i]} | y_{[1,i]}) \\ &= \pi(\theta, \lambda | \Delta n_{(0,i]}, \tilde{\beta}_{[0,i]}) \pi(\phi, \tilde{\beta}_{[0,i]}, \Delta n_{(0,i]} | y_{[1,i]})\end{aligned}$$

$$= \pi(\theta|\Delta n_{(0,i]})\pi(\lambda|\tilde{\beta}_{[0,i]})\pi(\phi, \tilde{\beta}_{[0,i]}, \Delta n_{(0,i]}|y_{[1,i]})$$

where the last two lines follow from the conditional independence relationships present in the model. Assuming an independent prior specification for the components of $\theta = (\kappa, \gamma)'$, with $\kappa \sim \text{Gamma}(a_\kappa, b_\kappa)$ and $\gamma \sim \text{Gamma}(a_\gamma, b_\gamma)$ gives

$$\kappa|\Delta n_{(0,i]} \sim \text{Gamma}\left(a_\kappa + \sum_{l=1}^{i}\sum_{j=1}^{m} \Delta n_{2,\tau_{l,j}}, b_\kappa + \sum_{l=1}^{i}\sum_{j=1}^{m} g_2(x_{\tau_{l,j-1}})\Delta\tau\right), \tag{9}$$

$$\gamma|\Delta n_{(0,i]} \sim \text{Gamma}\left(a_\gamma + \sum_{l=1}^{i}\sum_{j=1}^{m} \Delta n_{3,\tau_{l,j}}, b_\gamma + \sum_{l=1}^{i}\sum_{j=1}^{m} g_3(x_{\tau_{l,j-1}})\Delta\tau\right), \tag{10}$$

where $g_2(x) = h_2(x)/\kappa$ and $g_3(x) = h_3(x)/\gamma$, that is, the combinatorial factors in the hazard functions governing the infection and removal events. Similarly, if the discretised SDE (2) is adopted for the contact rate process $\tilde{\beta}_t$, then we may ascribe the prior $\lambda \sim \text{Gamma}(a_\lambda, b_\lambda)$ leading to the conditional posterior

$$\lambda|\tilde{\beta}_{[0,i]} \sim \text{Gamma}\left(a_\lambda + \frac{im}{2}, b_\lambda + \frac{1}{2}\sum_{l=1}^{i}\sum_{j=1}^{m}(\tilde{\beta}_{\tau_{l,j-1}} - \tilde{\beta}_{\tau_{l,j}})^2/\Delta\tau\right). \tag{11}$$

Hence, the conditional posterior $\pi(\theta, \lambda|\Delta n_{(0,i]}, \tilde{\beta}_{[0,i]})$ can be summarised by a vector of sufficient statistics $T_i := T_i(\Delta n_{(0,i]}, \tilde{\beta}_{[0,i]})$ given by the 3 shape and 3 rate hyper-parameters in (9)–(11). This vector can then be updated recursively within the particle filter, upon initialising with $T_0 = (a_\kappa, a_\gamma, a_\lambda, b_\kappa, b_\gamma, b_\lambda)$. In turn, particles $\theta^{(k)}$ and $\lambda^{(k)}$ can be updated by drawing from (9)–(11) conditional on $T_i^{(k)} := T_i\left(\Delta n_{(0,i]}^{(k)}, \tilde{\beta}_{[0,i]}^{(k)}\right)$.

Our treatment of ϕ depends on the observation model used. In the case of the Binomial model in (5), with $\phi = \rho$, the conditional posterior $\pi(\phi|\Delta n_{(0,i]}, y_{[1,i]})$ is tractable upon ascribing a $\text{Beta}(a_\phi, b_\phi)$ prior to ϕ. We obtain

$$\phi|\Delta n_{(0,i]}, y_{[1,i]} \sim \text{Beta}\left(a_\phi + \sum_{l=1}^{i} y_{t_l}, b_\phi + P'\sum_{l=1}^{i}\Delta N_{t_l} - \sum_{l=1}^{i} y_{t_l}\right).$$

Unfortunately, in the case of the Negative Binomial model (6) with $\phi = (\rho, \nu)'$, $\pi(\phi|\Delta n_{(0,i]}, y_{[1,i]})$ remains intractable. We therefore treat ϕ as a dynamic parameter and induce artificial evolution of ϕ by following the approach of Liu and West (2001). That is, if $\phi_{t_i}^{(k)}$ denotes the kth particle of ϕ at time t_i, we propagate forward to time t_{i+1} via

$$\phi_{t_{i+1}}^{(k)} \sim N\left(m_i^{(k)}, s^2 V_i\right)$$

where $m_i^{(k)} = a\phi_{t_i}^{(k)} + (1-a)\bar{\phi}_{t_i}$, $a = \sqrt{1-s^2}$, $V_i = \sum_{k=1}^{N}(\phi_{t_i}^{(k)} - \bar{\phi}_{t_i})(\phi_{t_i}^{(k)} - \bar{\phi}_{t_i})'/N$ and $\bar{\phi}_{t_i} = \sum_{k=1}^{N}\phi_{t_i}^{(k)}/N$. The practitioner can choose the hyper-parameters a and s to control shrinkage and over-dispersion. Guidance for choosing these hyper-parameters is

Algorithm 2 Particle Filter.

Input: data $y = (y_{t_1}, \ldots, y_{t_L})'$, number of particles N, initial draws $\psi^{(1:N)} \sim \pi(\psi)$, $x_{t_0}^{(1:N)} \sim \pi(x_{t_0})$, $\tilde{\beta}_{t_0}^{(1:N)} \sim \pi(\tilde{\beta}_{t_0})$ and sufficient statistic $T_0^{1:N}$.
For $i = 0, \ldots, L-1$ and $k = 1, \ldots, N$:

1. **Propagate** dynamic processes:
   a. draw $\phi_{t_{i+1}}^{(k)} \sim N(m_i^{(k)}, s^2 V_i)$ and set $\phi^{(k)} := \phi_{t_{i+1}}^{(k)}$,
   b. draw $\tilde{\beta}_{(i,i+1]}^{(k)} \sim \pi(\cdot|\tilde{\beta}_{t_i}^{(k)}, \lambda^{(k)})$,
   c. draw $\Delta n_{(i,i+1]}^{(k)} \sim q(\cdot|\tilde{\beta}_{[i,i+1)}^{(k)}, \phi_{t_{i+1}}^{(k)}, \theta^{(k)}, y_{t_{i+1}})$.
2. **Resample** with replacement among $\left(\psi^{(1:N)}, \tilde{\beta}_{t_{i+1}}^{(1:N)}, \Delta n_{t_{i+1}}^{(1:N)}\right)$ using the weights

$$w_{i+1}^{(k)} \propto \frac{\pi(\Delta n_{(i,i+1]}^{(k)}|\tilde{\beta}_{[i,i+1)}^{(k)}, \theta^{(k)})\pi(y_{t_{i+1}}|\Delta n_{t_{i+1}}^{(k)}, \phi_{t_{i+1}}^{(k)})}{q(\Delta n_{(i,i+1]}^{(k)}|\tilde{\beta}_{[i,i+1)}^{(k)}, \phi_{t_{i+1}}^{(k)}, \theta^{(k)}, y_{t_{i+1}})}$$

   as probabilities.
3. **Update** sufficient statistic $T_{i+1}^{(k)} := T_{i+1}(T_i^{(k)}, \Delta n_{(i,i+1]}^{(k)}, \tilde{\beta}_{(i,i+1]}^{(k)})$.
4. **Sample** $\theta^{(k)} \sim \pi(\theta|T_{i+1}^{(k)})$ and $\lambda^{(k)} \sim \pi(\lambda|T_{i+1}^{(k)})$ using (9)–(11).

Output: particle representation $\left(\psi^{(1:N)}, \tilde{\beta}_{t_i}^{(1:N)}, \Delta n_{t_i}^{(1:N)}\right)$ of the filtering distributions $\pi(\psi, \tilde{\beta}_{t_i}, \Delta n_{t_i}|y_{[1,i]})$, $i = 1, \ldots, L$.

given by Liu and West (2001); for example, we may specify $s^2 = 1 - ((3\delta - 1)/2\delta)^2$ for a discount factor δ typically taken to be 0.95-0.99.

We propagate the dynamic $\tilde{\beta}$ process blindly by drawing $\tilde{\beta}_{(i,i+1]}^{(k)} \sim \pi(\cdot|\tilde{\beta}_{t_i}^{(k)}, \lambda^{(k)})$. To mitigate against jump process trajectories being inconsistent with the next observation, we propagate $\Delta n_{(i,i+1]}^{(k)}$ conditional on $y_{t_{i+1}}$, by drawing

$$\Delta n_{(i,i+1]}^{(k)} \sim q\left(\cdot|\tilde{\beta}_{[i,i+1)}^{(k)}, \phi_{t_{i+1}}^{(k)}, \theta^{(k)}, y_{t_{i+1}}\right)$$

for some suitable construct $q\left(\cdot|\tilde{\beta}_{[i,i+1)}^{(k)}, \phi_{t_{i+1}}^{(k)}, \theta^{(k)}, y_{t_{i+1}}\right)$, discussion of which is deferred until Section 3.4. The complete particle filtering scheme (as appropriate for observation model (6)) is summarised in Algorithm 2.

3.4 Bridge construct

As previously discussed, we wish to propagate particle trajectories conditional on the next observation. To this end, we derive an approximate instantaneous rate or hazard function that is conditioned on the next observation. The derivation involves the construction of Gaussian approximation to the joint distribution of the incidence over a

time window whose right end-point is the next observation time, and the next observation itself. The conditioned hazard is then taken to be proportional to the expectation of the incidence given the observation.

Suppose we receive an observation y_{t_i} and have simulated as far as $\tau_{i,j} \in (t_{i-1}, t_i]$ so that the state $x_{\tau_{i,j}}$ and number of reaction events $\Delta n_{(t_{i-1},\tau_{i,j}]}$ are fixed and known. By analogy with the unconditioned hazard function, we denote the conditioned hazard function by $h^*\left(x_{\tau_{i,j}}|y_{t_i}\right)$, with dependence on the transmission model parameters θ, observation model parameters ϕ, time-varying contact process $\tilde{\beta}_{t_{i-1}}$ and incidence process $\Delta n_{(t_{i-1},\tau_{i,j}]}$ suppressed for notational simplicity.

Let $\Delta N_{(\tau_{i,j},t_i]}$ denote the number of reaction events in the interval $(\tau_{i,j}, t_i]$. The dSEIR model has the kth component of $\Delta N_{(\tau_{i,j},t_i]}$ distributed as a Poisson random variable with rate $h_k(x_{\tau_{i,j}})(t_i - \tau_{i,j})$. A Gaussian approximation of $\Delta N_{(\tau_{i,j},t_i]}$ then follows as

$$\Delta N_{(\tau_{i,j},t_i]} \overset{\text{approx.}}{\sim} \mathrm{N}\left\{h(x_{\tau_{i,j}})(t_i - \tau_{i,j}), H(x_{\tau_{i,j}})(t_i - \tau_{i,j})\right\} \tag{12}$$

where $H(x_{\tau_{i,j}}) = \mathrm{diag}\left\{h(x_{\tau_{i,j}})\right\}$. Moreover, taking a Gaussian approximation of the Binomial observation model given by (5) and Negative Binomial observation model given by (6), leads to the distributional form

$$Y_{t_i}|\Delta N_{(\tau_{i,j},t_i]} \overset{\text{approx.}}{\sim} \mathrm{N}\left\{\mu(\Delta n_{(t_{i-1},\tau_{i,j}]}), \sigma^2(\Delta n_{(t_{i-1},\tau_{i,j}]})\right\}. \tag{13}$$

Here, $\mu(\Delta n_{(t_{i-1},\tau_{i,j}]}) = \rho P'(\Delta n_{(t_{i-1},\tau_{i,j}]} + \Delta N_{(\tau_{i,j},t_i]})$ for both Binomial and Negative Binomial observation models, with dependence on $\Delta N_{(\tau_{i,j},t_i]}$ suppressed for notational ease. The variance term is model dependent, taking the form

$$\sigma^2(\Delta n_{(t_{i-1},\tau_{i,j}]}) = (1-\rho)\mu(\Delta n_{(t_{i-1},\tau_{i,j}]}) \tag{14}$$

in the case of a Binomial observation model and

$$\sigma^2(\Delta n_{(t_{i-1},\tau_{i,j}]}) = \mu(\Delta n_{(t_{i-1},\tau_{i,j}]}) + \mu(\Delta n_{(t_{i-1},\tau_{i,j}]})^2/\nu \tag{15}$$

in the case of a Negative Binomial observation model. Note that (14) and (15) depend on $\Delta N_{(\tau_{i,j},t_i]}$. To force a joint Gaussian distribution of $\Delta N_{(\tau_{i,j},t_i]}$ and Y_{t_i}, we replace $\mu(\Delta n_{(t_{i-1},\tau_{i,j}]})$ in (14) and (15) with $\hat{\mu}(\Delta n_{(t_{i-1},\tau_{i,j}]}) = \rho P'(\Delta n_{(t_{i-1},\tau_{i,j}]} + \Delta \hat{N}_{(\tau_{i,j},t_i]})$ where $\Delta \hat{N}_{(\tau_{i,j},t_i]}$ is the mean of the approximating Normal distribution in (12). Then, marginalising $\Delta N_{(\tau_{i,j},t_i]}$ in (13) through the use of (12) gives

$$Y_{t_i} \overset{\text{approx.}}{\sim} \mathrm{N}\left\{\hat{\mu}(\Delta n_{(t_{i-1},\tau_{i,j}]}), \rho^2 P' H(x_{\tau_{i,j}}) P(t_i - \tau_{i,j}) + \sigma^2(\Delta n_{(t_{i-1},\tau_{i,j}]})\right\}. \tag{16}$$

Similarly, we use (12) and (13) to give the covariance between Y_{t_i} and $\Delta N_{(\tau_{i,j},t_i]}$ as

$$\begin{aligned}\mathrm{Cov}(Y_{t_i}, \Delta N_{(\tau_{i,j},t_i]}) &= \mathrm{Cov}(\rho P' \Delta N_{(\tau_{i,j},t_i]}, \Delta N_{(\tau_{i,j},t_i]}) \\ &= \rho P' H(x_{\tau_{i,j}})(t_i - \tau_{i,j}).\end{aligned} \tag{17}$$

The approximate joint distribution of $\Delta N_{(\tau_{i,j},t_i]}$ and Y_{t_i} is then given through combination of (12), (13) and (17). This is

$$\begin{pmatrix} \Delta N_{(\tau_{i,j},t_i]} \\ Y_{t_i} \end{pmatrix} \overset{\text{approx.}}{\sim} \mathrm{N}\left\{ \begin{pmatrix} h(x_{\tau_{i,j}})(t_i - \tau_{i,j}) \\ \hat{\mu}(\Delta n_{(t_{i-1},\tau_{i,j}]}) \end{pmatrix}, \right.$$
$$\left. \begin{pmatrix} H(x_{\tau_{i,j}})(t_i - \tau_{i,j}) & \rho H(x_{\tau_{i,j}})P(t_i - \tau_{i,j}) \\ \rho P' H(x_{\tau_{i,j}})(t_i - \tau_{i,j}) & \rho^2 P' H(x_{\tau_{i,j}})P(t_i - \tau_{i,j}) + \sigma^2(\Delta n_{(t_{i-1},\tau_{i,j}]}) \end{pmatrix} \right\}. \quad (18)$$

Finally, we condition on $Y_{t_i} = y_{t_i}$, take the expectation of the resulting distribution and divide by $(t_i - \tau_{i,j})$ to obtain the conditioned hazard function as

$$h^*(x_{\tau_{i,j}}|y_{t_i}) = h(x_{\tau_{i,j}}) + \rho P' H(x_{\tau_{i,j}})\{\rho^2 P' H(x_{\tau_{i,j}})P(t_i - \tau_{i,j}) + \sigma^2(\Delta n_{(t_{i-1},\tau_{i,j}]})\}^{-1}$$
$$\times \{y_{t_i} - \hat{\mu}(\Delta n_{(t_{i-1},\tau_{i,j}]})\}.$$

It is then straightforward to generate end-point conditioned trajectories, for example, over an interval $(t_{i-1}, t_i]$ by executing Algorithm 1, with h replaced by h^* and time step $\Delta\tau$. Moreover, the corresponding likelihood $q(\cdot|\tilde{\beta}_{[i-1,i)}, \phi_{t_i}, \theta, y_{t_i})$ is simply a product of Poisson probability mass functions, each with rate $h^*(x_{\tau_{i,j}}|y_{t_i})\Delta\tau$, for $j = 0, \ldots, m-1$.

4 Applications

Here, we consider three different applications of the particle filter (Algorithm 2) using both synthetic and real data to demonstrate the effectiveness of the scheme. The first application uses synthetic data generated using the SIR model. We compare the accuracy of the resulting marginal parameter posteriors with a pseudo-marginal Metropolis-Hastings scheme. The other two applications are real data examples, taken from Fintzi et al. (2022) and Spannaus et al. (2022), respectively. In all cases, we set the time-step of the dS(E)IR model to be $\Delta\tau = 0.1$ (in units of weeks for the two real data applications), which gave a reasonable balance between accuracy and computational cost. All algorithms were coded in R and run on a PC with a 2.6 GHz clock speed across 20 cores. Computer code to reproduce the following experiments can be downloaded from `https://github.com/Sam-Whitaker/Sequential_Bayes_Epi`.

4.1 Simulation study

We consider the dSIR model and, initially, a single synthetic data set consisting of the (noisy) number of new infections in $(t-1, t]$ for $t = 1, \ldots, 10$. The data were generated by applying Algorithm 1 with a time step of 0.001 over the time interval $[0, 10]$ and summing the number of infections over equally spaced sub-intervals of unit length. To emulate the influenza data set described in BMJ News and Notes (1978), we took the initial state to be $(s_0, i_0)' = (762, 5)'$, a removal rate of $\gamma = 0.5$ and assumed a time-varying infection rate whose logarithm is of the form

$$\mathrm{d}\tilde{\beta}_t = \lambda^{-1/2}\mathrm{d}W_t, \quad \tilde{\beta}_0 = -6 \quad \text{and} \quad \lambda = 100$$

so that the infection rate itself is scaled Brownian motion. Finally, we corrupted the data via a Binomial observation model with parameter $\rho = 0.9$. The data are shown in Figure 1.

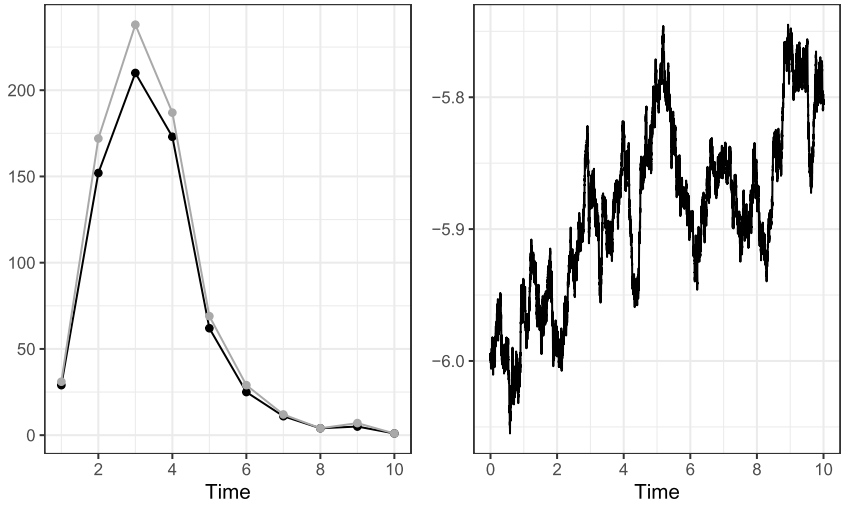


Figure 1: Synthetic data application. Left panel: observed new infections (black) and underlying latent infections (grey) in $(t-1, t]$ for $t = 1, \ldots, 10$. Right panel: time-varying (log) infection rate $\tilde{\beta}_t$.

We adopted a weakly informative prior specification by taking $\log(\beta_0) \sim \mathrm{N}(-6.5, 0.5^2)$, $\gamma \sim \mathrm{Gamma}(11, 20)$, $\lambda \sim \mathrm{Gamma}(15, 0.14)$ and $\rho \sim \mathrm{Beta}(90, 15)$. The particle filter was run with $N = 10^k$ particles, for $k = 1, 2, \ldots, 6$. The wall clock computing time versus N is shown in Figure 2 for serial and parallel implementations; a single run with $N = 10^6$ takes approximately 16 minutes. The benefit of parallelising the particle filter is clear, with an order of magnitude speed-up (over a serial implementation) achieved for $N \geq 10^4$ particles.

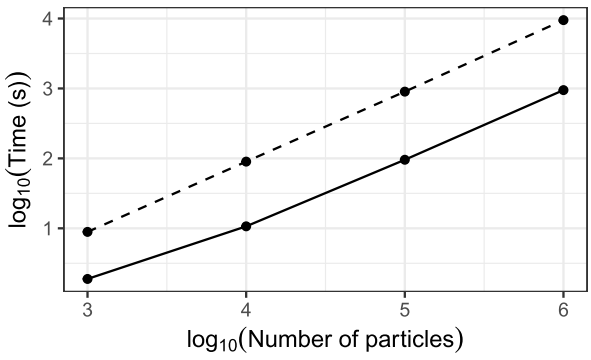


Figure 2: Synthetic data application. Wall clock computing times (in seconds) for runs of the scheme in serial (dashed) and parallel (solid) versus the number of particles.

To benchmark accuracy for different values of N, we generated samples from the marginal parameter posteriors via a long run (10^6 iterations) of a pseudo-marginal Metropolis-Hastings scheme (PMMH, e.g. Andrieu et al., 2010) targeting the posterior under the dSIR model. In more detail, we applied a Gaussian random walk proposal to the natural logarithm of the model components; the innovation variance was tuned (from a short pilot run) to meet a desired acceptance rate (see e.g. Schmon et al., 2021). The observed-data likelihood was (unbiasedly) estimated using a bootstrap particle filter over the latent state process, with particles determined via the practical advice given in Sherlock et al. (2015).

For each component of the parameter vector $\psi = (\gamma, \lambda, \rho)'$ we computed "gold standard" estimates of the marginal posterior expectations $\mathrm{E}(\psi_i|y)$ and standard deviations $\mathrm{SD}(\psi_i|y)$, denoted by e_i and s_i, respectively. Replicate runs of the particle filter are then used to obtain corresponding estimates $\hat{e}_i^{(N,j)}$ and $\hat{s}_i^{(N,j)}$ where N denotes the number of particles used and $j = 1, \ldots, R$ indexes the replicate run. Bias and root mean squared error (RMSE) of the particle filter's estimator of e_i is then calculated as

$$\mathrm{Bias}(\hat{e}_i^{(N)}) = \frac{1}{R}\sum_{j=1}^{R}(\hat{e}_i^{(N,j)} - e_i) \quad \text{and} \quad \mathrm{RMSE}(\hat{e}_i^{(N)}) = \sqrt{\frac{1}{R}\sum_{j=1}^{R}(\hat{e}_i^{(N,j)} - e_i)^2}$$

with similar expressions obtained for the estimator of s_i. Bias and RMSE for the estimator of the marginal posterior expectation and standard deviation of γ (removal rate) and ρ (reporting rate) can be found in Figure 3. Unsurprisingly, Bias and RMSE reduce as the number of particles increases. There is relatively little improvement in accuracy beyond $N = 50K$, suggesting that for this scenario, this value of N gives a reasonable balance between accuracy and efficiency. This remark is further supported by Figure 4, which shows the particle filter sample output (after assimilating all observations) for $N = 50K$, with the gold standard PMMH marginal posterior densities overlaid.

Summaries (mean and 95% credible interval) of the marginal filtering distributions $\pi(\psi|y_{[1,t]})$ and $\pi(\beta_t|y_{[1,t]})$ are shown in Figure 5. Note that corresponding summaries of the prior distribution are shown at time $t = 0$. Samples from the filtering distributions are consistent with the ground truth parameter values that produced the data. Posterior uncertainty generally reduces as more data points are assimilated. However, there is little difference between the prior distribution of the precision parameter λ governing the time-varying infection rate and the filtering distribution of λ at each observation time-point; we anticipate that this parameter will be particularly sensitive to the choice of prior.

The particle filter gives samples of incidence counts over $(t_{i-1}, t_i]$, denoted $\Delta n_{t_i}^{(1:N)}$, from the (marginal) filtering distributions $\pi(\Delta n_{t_i}|y_{[1,i]})$, $i = 1, \ldots, 10$. Given corresponding samples of $x_{t_{i-1}}^{(1:N)}$, the prevalence at time t_i can be computed for each sample k via

$$x_{t_i}^{(k)} = x_{t_{i-1}}^{(k)} + \sum_j A_j' \Delta n_{j,t_i}^{(k)}.$$

Hence, samples from the filtering distributions for prevalence, $\pi(x_{t_i}|y_{[1,i]})$, are easily generated as part of the particle filter. Summaries of these filtering distributions are

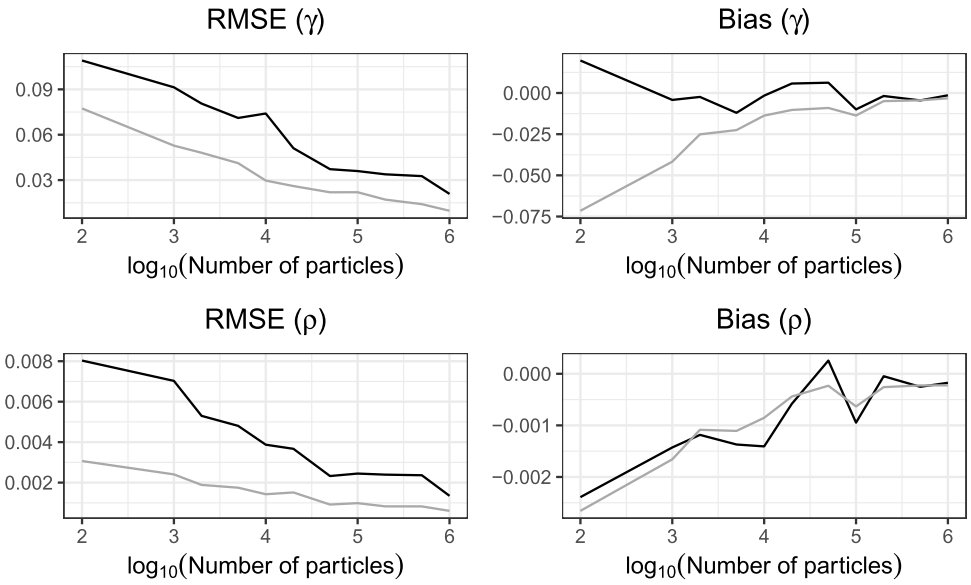


Figure 3: Synthetic data application. RMSE (left) and bias (right) versus $\log_{10}(N)$ for the particle filter estimator of $e_1 = \mathrm{E}(\gamma|y)$ (black, top row) and $s_1 = \mathrm{SD}(\gamma|y)$ (grey, top row) and $e_3 = \mathrm{E}(\rho|y)$ (black, bottom row) and $s_1 = \mathrm{SD}(\rho|y)$ (grey, bottom row).

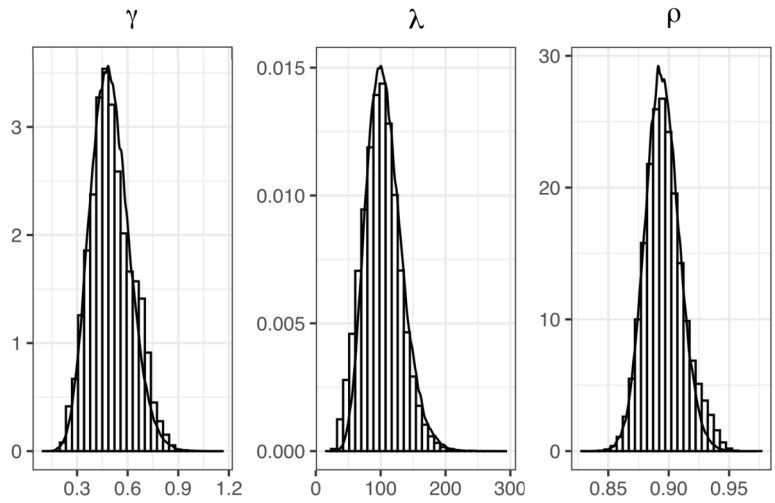


Figure 4: Synthetic data application. Posterior output from the particle filter (histograms) and ground truth posterior densities (kernel density estimates overlaid) obtained via PMMH (10^6 iterations).

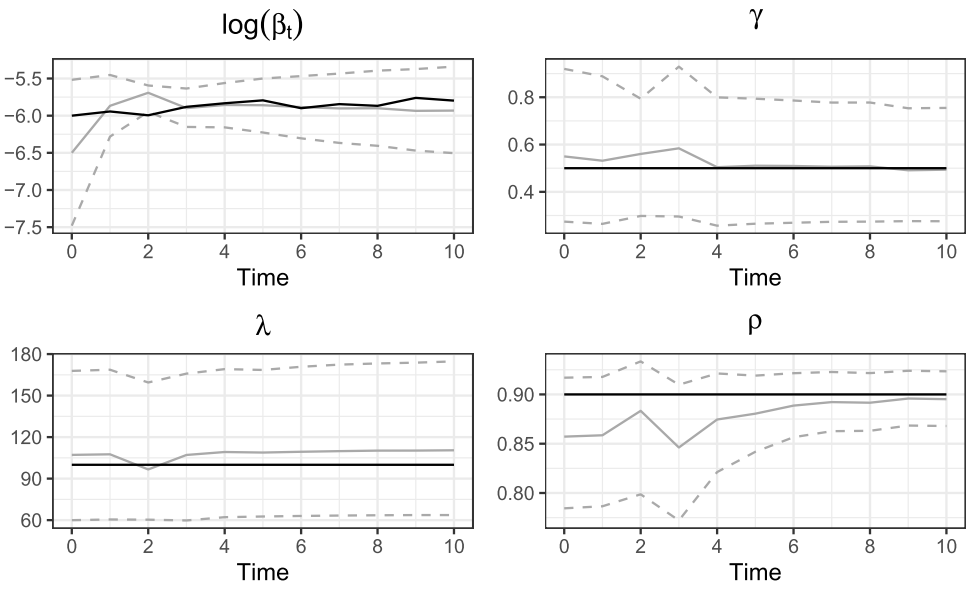


Figure 5: Synthetic data application. Filtering mean (solid grey) and 95% interval (dashed grey) of the latent infection rate (top left), removal rate (top right), infection precision (bottom left) and reporting rate (bottom right). The ground truth is indicated (black).

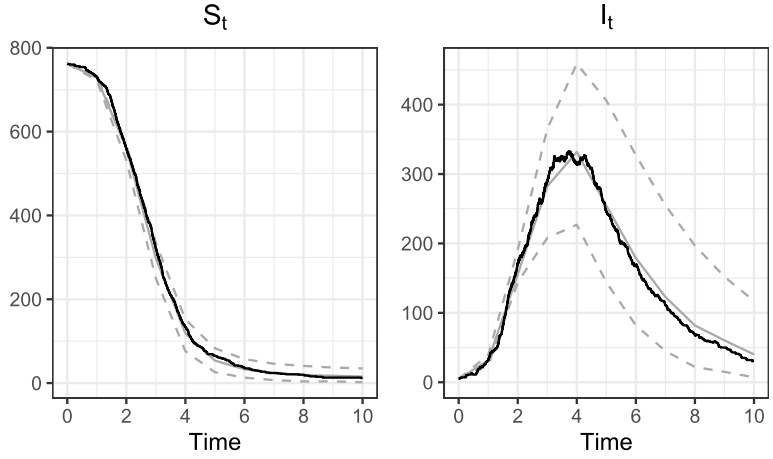


Figure 6: Synthetic data application. Filtering mean (solid grey) and 95% credible interval (dashed grey) for the susceptible (left) and infective (right) species. The ground truth is shown in black.

shown in Figure 6, from which we see that the filter output is consistent with the ground truth.

4.2 Ebola

We consider the first of two real data examples by applying the proposed dSEIR model and inference scheme to a subset of the data found in Fintzi et al. (2022). In particular, we only consider data from Sierra Leone. The data are 53 weekly observations of the incidence of Ebola from May 2014 to May 2015; these are shown in Figure 7.

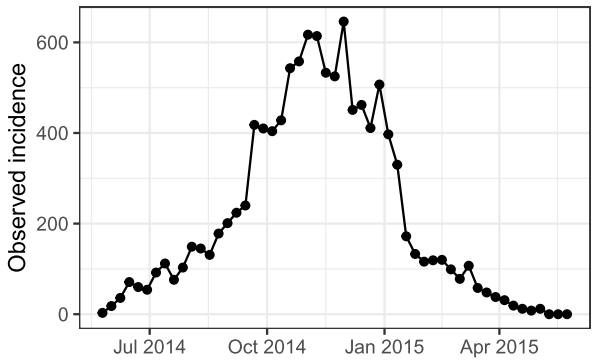


Figure 7: Ebola application. Weekly incidence data from the Ebola outbreak in Sierra Leone. Tick marks correspond to time intervals of length three months.

Following Fintzi et al. (2022), we specify a Negative Binomial observation model to link the true unobserved incidence process to observations; see Equation (6). The authors include a constant immigration of cases, occurring with rate α, via a modification of the hazard of the first reaction, $h_1(x_t)$. As we are only considering one country in isolation, we ignore this proposed immigration, assuming that the import and export rates will be approximately equal. Instead, we fit the dSEIR model with a *constant* contact rate β, as described in Section 2.

Our choice of prior is consistent with the specification used in Fintzi et al. (2022); we take $\beta \sim \text{Gamma}(2, 50000)$, $\kappa \sim \text{Gamma}(5, 4.6)$, $\gamma \sim \text{Gamma}(10, 10)$, $\text{logit}(\rho) \sim \text{N}(0.85, 0.75^2)$ and $\nu \sim \text{Gamma}(5, 0.2)$. We assume the initial state is fixed and known to be $x_0 = (44326, 15, 10)'$. We found that taking the number of particles to be at least $N = 4 \times 10^6$ is necessary to avoid degeneracy in the parameter samples governing the observation model (for which we resort to the jittering approach described in Section 3.3). A single run of Algorithm 2 (parallelised over 20 cores) with this choice of N took approximately 321 minutes (or around 5.3 hours).

As in Section 4.1, we provide summaries (mean and 95% credible interval) of the marginal filtering distributions of each parameter in Figure 8. We see that the uncertainty reduces from prior to posterior for most parameters, with the exception of the reporting rate ρ, which remains almost unchanged. This reduction in uncertainty is most prominent for both the contact rate, β and the over-dispersion parameter ν.

Given in Figure 9 are summaries of the filtering distributions for the prevalence. We see that the credible region for the susceptible species narrows towards the middle of

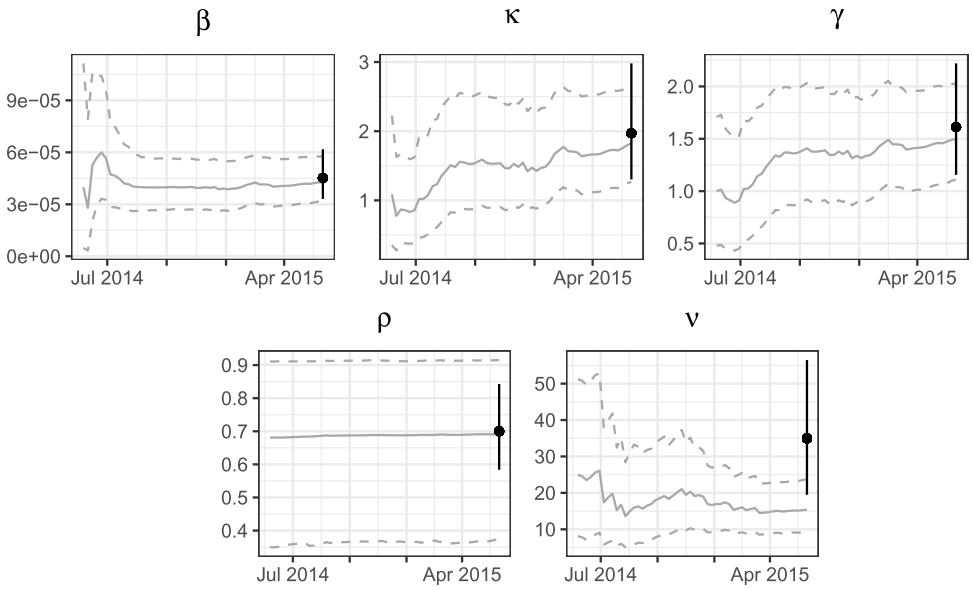


Figure 8: Ebola application. Filtering mean (solid grey) and 95% credible interval (dashed grey) of the contact rate (top left), infection rate (top centre), removal rate (top right), reporting rate (bottom left) and overdispersion parameter (bottom right) for the dSEIR model. Overlaid are filtering means (black dots) and 95% credible intervals (black lines) at the final time, using the linear noise approximation (LNA) as the transmission model.

the data (around Nov-Dec 2014) before widening towards the end again. Conversely, the uncertainty increases towards the middle of the data in both the exposed and infectious species. The peaks in both the exposed and infectious species correspond to peaks in the data, which is to be expected; as the population of the exposed/infectious species increases, so will the weekly incidence.

Finally, we produce five one-step-ahead forecasts of the weekly incidence for the last five non-zero observations in the data. To produce a forecast for the observed weekly incidence at an arbitrary time T, we run Algorithm 2 up to time $T-1$, and then, using the particle representation of $\pi(\phi, \tilde{\beta}_{[0,T-1]}, \Delta n_{(0,T-1]} | y_{[1,T-1]})$ from the output of Algorithm 2, we generate N realisations of the incidence count over $(T-1, T]$. These simulated incidence counts are subjected to noise as per the observation model, and the resulting samples are then summarised using a box and whisker plot. We repeat this process for each of five forecast intervals of interest. The resulting box and whisker plots are provided in Figure 10 (left panel). We see that the observed values, given as dashed lines, lie close to the median value for each of the five forecasts and fall between the lower and upper quartiles, suggesting that the dSEIR model (in conjunction with the proposed inference scheme) is able to accurately forecast Ebola incidence.

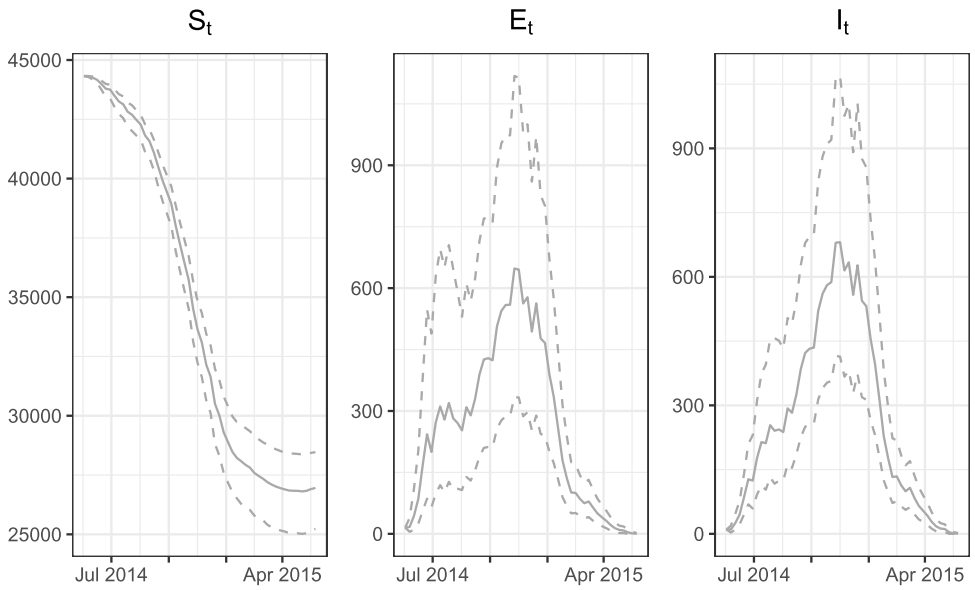


Figure 9: Ebola application. Filtering mean (solid grey) and 95% credible interval (dashed grey) for the susceptible, exposed and infective species under the dSEIR model.

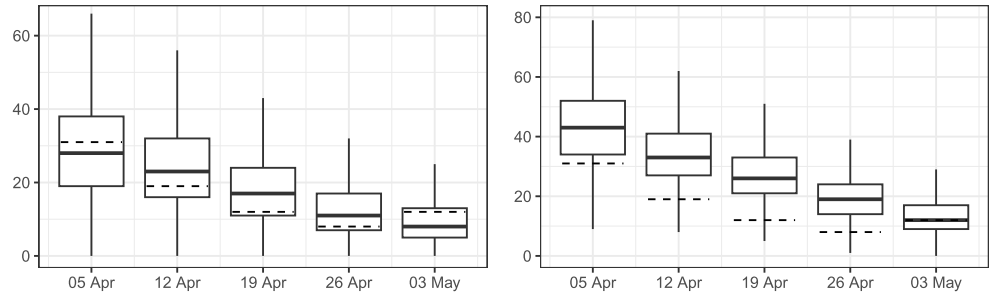


Figure 10: Ebola application. Five one step ahead dSEIR forecasts (left panel) and LNA forecasts (right panel) of the final five (non-zero) observed incidences overlaid with the true observed values (dashed).

Comparison with a linear noise approximation (LNA)

Here, we compare forecast accuracy of the dSEIR model to the approach taken by Fintzi et al. (2022), who describe SEIR dynamics using a linear noise approximation (see e.g. Komorowski et al., 2009; Fearnhead et al., 2014; Schnoerr et al., 2017). That is, the LNA is used as the transmission model and the previously described Negative Binomial model is used as the observation model. In order to generate samples from the marginal parameter posterior, an MCMC scheme is used. In particular, and for reasons

of computational efficiency, we implement the marginal Metropolis-Hastings scheme of Golightly et al. (2023); brief details regarding the LNA and the MCMC scheme used in this setting are provided in the supplementary material (Whitaker et al., 2025).

The LNA requires solution of a coupled ordinary differential equation (ODE) system. We used a simple Euler solver with a time step of 0.01, which gave a reasonable computational efficiency and accuracy balance. The prior specification is as previously described. We ran the MCMC scheme for 100k iterations, using a correlated random walk proposal, tuned using the posterior variance from a shorter pilot run of 10K iterations. The resulting computational cost is approximately equal (in terms of CPU wall clock time) to the particle filter run with $N = 4 \times 10^6$.

Posterior means and 95% credible intervals for the model parameters under the LNA are compared to the same posterior summaries obtained from the particle filter applied to the dSEIR model in Figure 8. It is evident that posterior summaries under the LNA are mostly consistent with those obtained under the dSEIR model, with the exception of the overdispersion parameter, ν, whose posterior mass is concentrated around larger values than that from the particle filter. There is a clear reduction in posterior variance for the reporting rate under the LNA compared to dSEIR, although the mean from both approaches appears consistent.

As a final comparison, we now generate five one-step-ahead forecasts of the last five non-zero observations under the LNA. To generate a forecast at time T, we ran the LNA MCMC scheme using all the data up to and including time $T-1$, storing each accepted parameter vector along with a corresponding draw from $N_{T-1}|Y_{1:T-1}$. We then propagate a thinned sample forward over the interval $(T-1, T)$ to obtain approximate incidence counts at time T, which we subjected to noise according to the observation model. The forecasts are summarised by box and whisker plots in Figure 10 (right panel). A comparison of forecast distributions in Figure 10 suggests that the LNA tends to overestimate the observations, with four out of five median forecasts lying below the lower quartile. Nevertheless, the forecast uncertainty is comparable between models; when using the LNA, relatively larger values of the overdispersion parameter ν are inferred (see Figure 8), seemingly compensating for the relative increase in the observation mean. Although we do not pursue it here, deriving the LNA from a log transformed SDE approximation to the MJP (rather than on the natural scale) may account for the discrepancy in forecast accuracy. We refer the reader to Fintzi et al. (2022) for further details of this approach.

4.3 Covid-19 in New York

For the final real data application, we consider a data set taken from Spannaus et al. (2022). The data consist of twenty-five observations of the weekly incidence of COVID-19 in New York, starting in March 2020 and ending in mid-August 2020. The data are presented in Figure 11.

In Spannaus et al. (2022), both SIR and SEIR models are considered. For simplicity, we choose to adopt the former, and model the data using an SIR model with dynamic contact and reporting rates, and a Negative Binomial observation model. Specifically,

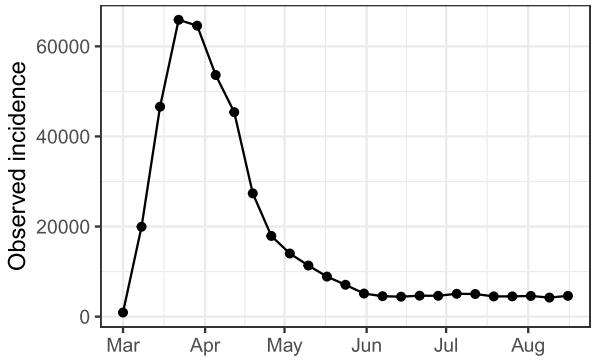


Figure 11: COVID-19 application. Weekly incidence in New York.

we allow the logit of the reporting rate to vary through time according to a scaled Brownian motion process whose precision is given by λ_ρ. To avoid any ambiguity, we denote the precision for the scaled Brownian motion process governing the log of the contact rate by λ_β. Hence, the dSIR transmission model is

$$\begin{cases} X_{t+\Delta t} = x_t + A'\Delta N_t, \\ \log \beta_{t+\Delta t} = \log \beta_t + \lambda_\beta^{-1/2} \Delta W_{1,t}, \\ \text{logit}\rho_{t+\Delta t} = \text{logit}\rho_t + \lambda_\rho^{-1/2} \Delta W_{2,t}. \end{cases} \tag{19}$$

where $W_{1,t}$ and $W_{2,t}$ are uncorrelated Brownian motion processes. The observation model is

$$Y_{t_i} | \Delta N_{t_i} \sim \text{NegBin}(\mu_i = \rho_{t_i} P' \Delta N_{t_i}, \sigma_i^2 = \mu_i + \mu_i^2/\nu), \quad i = 1, \ldots, L. \tag{20}$$

We follow Spannaus et al. (2022) by adopting a prior specification as follows. We let $\log(\beta_0) = 0$, $\gamma \sim \text{Gamma}(11.088, 2.192)$, $\lambda_\beta \sim \text{Gamma}(4, 1)$, $\lambda_\rho \sim \text{Gamma}(4, 1)$, $\rho_0 \sim \text{Beta}(3, 2)$ and $1/\sqrt{\nu} \sim \text{U}(0, 0.5)$ *a priori*. We found that taking the number of particles to be at least $N = 4 \times 10^6$ avoids sample impoverishment. A single run of Algorithm 2 (parallelised over 20 cores) with this choice of N took approximately 49 minutes.

Summaries of the filtering distributions of the parameters and prevalence process are given in Figures 12 and 13. We can see that the uncertainty about the parameters reduces from prior to posterior in all cases, indicating that the analysis is informative.

Finally, we assess forecast accuracy of the dSIR model by providing five one-step-ahead forecasts for the final five non-zero observations, using the Monte Carlo method described in Section 4.2; see Figure 14 (left panel). It is evident that all forecast medians are consistent with the observations, suggesting good forecast accuracy. Although forecast uncertainty is relatively large (with interquartile ranges of around 7,500 cases), the forecast for the final observation is comparable to that given in Spannaus et al. (2022) (see Figure 5 therein). Further comparisons with the transmission model used in the latter are considered in the next section.

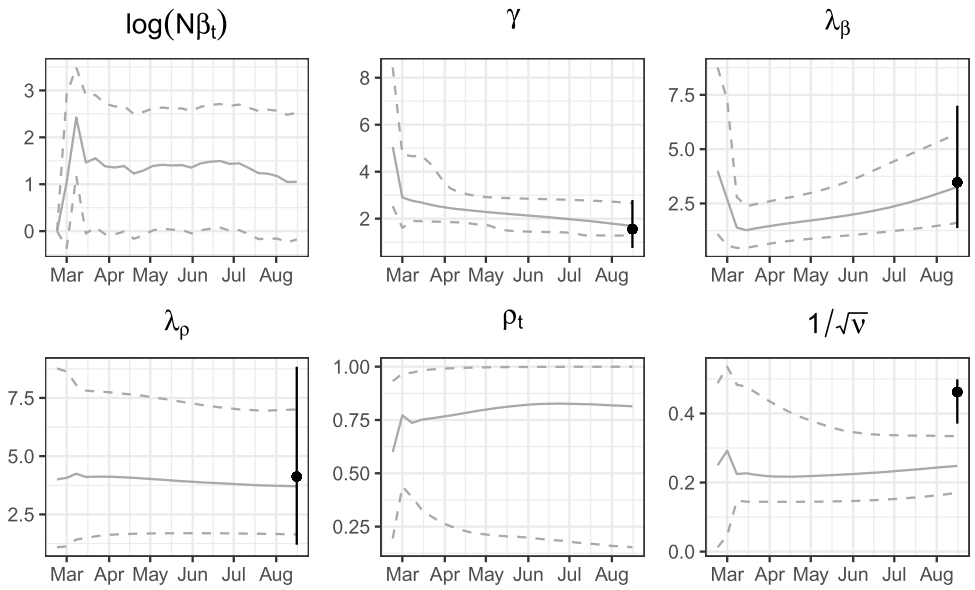


Figure 12: COVID-19 application. Mean (solid grey) and 95% interval (dashed grey) from the filtering distributions of $\log(N\beta_t)$ (top left), γ (top centre), λ_β (top right), λ_ρ (bottom left), ρ_t (bottom centre) and $1/\sqrt{\nu}$ (bottom right) under the dSIR model. Overlaid for the static parameters are filtering means (black dots) and 95% credible intervals (black lines) at the final time, using the transmission model of Spannaus et al.

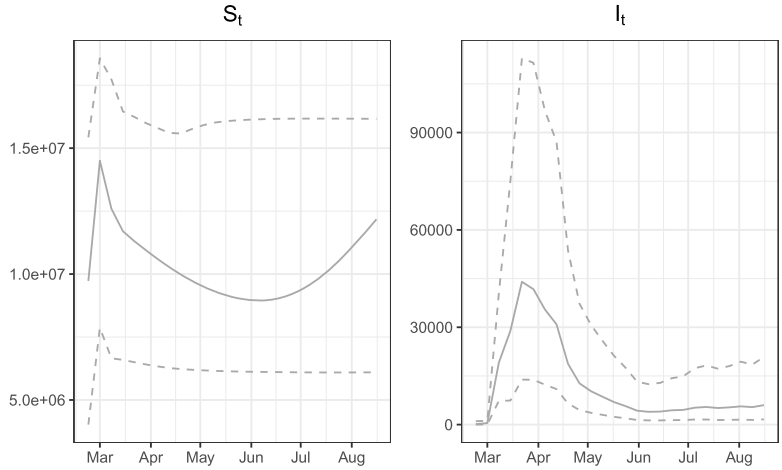


Figure 13: COVID-19 application. Mean (solid grey) and 95% interval (dashed grey) for the filtering distributions of the Susceptible (left) and Infective (right) species.

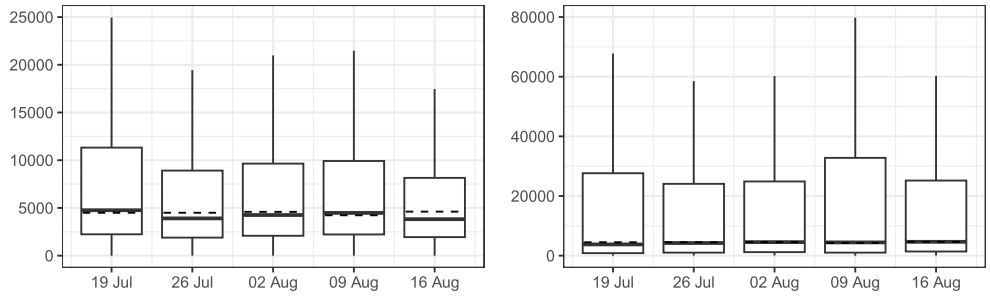


Figure 14: COVID-19 application. Five one-step-ahead forecasts under dSIR (left panel) and the model of Spannaus et al (right panel), of the final five observed incidences. The true observed values are overlaid (dashed).

Comparison with Spannaus et al.

The SIR transmission model in Spannaus et al. (2022) takes the form of the well studied Kermack-McKendrick ordinary differential equation (ODE) system, albeit with a time-varying infection rate. That is

$$\begin{cases} \mathrm{d}S_t = -\beta_t S_t I_t \mathrm{d}t, \\ \mathrm{d}I_t = (\beta_t S_t I_t - \gamma I_t)\mathrm{d}t, \\ \mathrm{d}R_t = \gamma I_t \mathrm{d}t, \end{cases} \tag{21}$$

where $\log \beta_t$ follows an SDE with solution as in (19). To facilitate comparison with the dSIR model, we assume that the logit reporting rate $\mathrm{logit}\rho_t$ follows an SDE with solution as in (19) and a Negative Binomial observation model as in (20).

To generate samples from the marginal parameter posterior under the assumption of (21), we ran the particle marginal Metropolis-Hastings (PMMH) scheme described in Spannaus et al. (2022) for 10^5 iterations. A Gaussian random walk proposal was applied to the natural logarithm of the model components, and the innovation variance and number of particles were optimised as per Section 4.1, resulting in 100 state particles. The wall clock time for this PMMH run was approximately 41 hours. For comparison, inference for the dSIR model using our proposed sequential inference scheme took approximately 50 minutes. We did not pursue a parallel implementation of PMMH due to the relatively small number of required state particles, which limits the potential gains in efficiency of a parallelised approach.

We report the posterior mean and 95% credible interval for each of the static parameters in Figure 12. It is clear that parameter samples are consistent with those obtained under the dSIR model (via the particle filter) for γ, λ_β and λ_ρ. However, for the over-dispersion parameter ν, there is some disagreement between the two marginal posterior distributions, with samples of $1/\sqrt{\nu}$ typically smaller under dSIR than when using the transmission model in (21).

We also provide five one-step-ahead forecasts under the assumption of the transmission model in (21), for the final five non-zero observations in the data set in Figure 14

(right panel). Comparing forecasts in Figure 14, we see agreement between forecast medians and the observations (for both schemes), although use of (21) as the inferential and forecasting model appears to result in increased forecast uncertainty (relative to dSIR). Given that broadly consistent inferences are obtained across models for the parameters governing the transmission process, it appears that the increased forecast uncertainty is due to the increased values of typical $1/\sqrt{\nu}$ samples, in turn resulting in a larger observation variance. We expect that the increased observation variance is essentially compensating for the assumption of deterministic dynamics in (21).

5 Summary

In this paper, we have considered the challenging problem of performing sequential Bayesian inference for the static parameters and latent, dynamic components of a stochastic representation of an epidemic compartment model. The inference task was developed under the assumption that only imperfect, discrete-time incidence values are available.

The most natural continuous-time Markov jump process (MJP) representation of cumulative incidence was replaced by a discrete-time approximation in which the numbers of each transition event (e.g. contact, infection or removal in an SEIR model) are assumed to be Poisson distributed over a time interval whose length is specified by the practitioner. The model allows for the straightforward incorporation of time-varying parameters via time-discretised stochastic differential equations (SDEs). The resulting dSEIR model admits several inferential advantages over other (stochastic) approximations to the MJP. Conditional on complete information regarding the cumulative incidence process, the static parameters governing the latent transmission model, and in some cases the observation model, follow a tractable posterior distribution. We leveraged this tractability via a sequential Monte Carlo scheme (particle filter) that rejuvenates parameter samples by drawing from their conditional parameter posterior after propagating and resampling particle trajectories of the latent incidence process. This approach essentially follows the framework of Storvik (2002) (see also Carvalho et al., 2010, for a related particle learning approach). We further modified the particle filter by propagating particle trajectories conditional on the next observation, by adapting the conditioned hazard of Golightly and Wilkinson (2015) to the incidence setting.

We applied the resulting methodology to synthetic and real data examples, benchmarking against competing models and inference schemes. The proposed particle filter benefits from a parallel implementation, as demonstrated using synthetic data generated from a stochastic SIR model with a time-varying infection rate and Binomial observation model. We observed with an order of magnitude speed-up (over a serial implementation) for $N \geq 10^4$ particles. We also achieve satisfactory inferential accuracy, as measured by bias and RMSE for key posterior quantities, relative to corresponding values obtained from long runs of a pseudo-marginal Metropolis-Hastings scheme. For the synthetic data application, these quantities stabilise with $N \geq 10^5$ particles. We compared forecast accuracy of the dS(E)IR model to two competing approaches via real data applications involving the spread of Ebola in West Africa (Fintzi et al.,

2022) and COVID-19 in New York (Spannaus et al., 2022). In the former, a linear noise approximation was used as a tractable approximation to the MJP representation of cumulative incidence. For a fixed computational budget, the dSEIR model better reflects the end of the epidemic, likely due to relatively low numbers of incidence, for which a continuous-valued approximation is likely to give a relatively poor approximation. For the final application involving COVID-19, we compared against a transmission model in which the infection rate of a deterministic ordinary differential equation (ODE) system is replaced with the solution of an SDE. We found that this approach leads to accurate point-summary forecasts but over-estimates uncertainty (relative to the dSIR model). It is also worth noting that both real data examples required an order of magnitude increase in the number of particles, relative to the synthetic data application. This is likely due to the relative increase in the number of observations and model components in the Ebola application, and the myopic propagation of the latent reporting process in the final application. Nevertheless, our empirical findings suggest that use of the particle filter allows for assimilating a new observation in around 6 minutes for the dSEIR model and 2 minutes for the simpler dSIR model.

5.1 Limitations

Here, we consider several limitations of the proposed approach and make some suggestions for future work.

The dS(E)IR model requires specification of a step size, which balances accuracy (relative to the most natural MJP representation of cumulative incidence) and computational cost. We have assumed a fixed step size but note that an optimal choice is likely to depend on the phase of the epidemic. Conditions under which the Poisson approximation is reasonable have been discussed by Gillespie, D. T. (2001) among others; for methods that allow a step size to be chosen adaptively, see e.g. Cao et al. (2006); Sandmann (2009).

The particle filter used in this paper propagates particles conditional on the next observation by replacing the Poisson rate with a hazard function derived via a Gaussian approximation of the cumulative incidence between the current time and next observation, conditional on the observation itself. A further assumption in deriving the resulting conditioned hazard is linearity of the latent process over a time interval of length at most given by the observation interval. In scenarios in which the epidemic is sparsely observed in time, to the extent that cumulative incidence does not evolve in a linear fashion, the conditioned hazard is likely to result in trajectories that are inconsistent with the true conditioned process. Recent work on sampling conditioned jump processes (in a chemical kinetics setting) is relevant here (see e.g. Golightly and Sherlock, 2019; Corstanje et al., 2023) and could be adapted to dS(E)IR model.

A key feature of the proposed inference scheme is the leveraging of a tractable conditional parameter posterior, which depends (given complete information on the latent process up to time t_i) on a low-dimensional sufficient statistic T_i. This approach also forms the basis of particle learning (Carvalho et al., 2010). Particle degeneracy in the latter suite of algorithms has been well documented (see e.g. Section 4 of Chopin et al.,

2010). That is, the number of different values of the statistic at a fixed time point prior to t_i that contribute to T_i is decreasing in i at an exponential rate. The applications presented here involve relatively short time-series coupled with a parallel implementation that allows sufficiently many particles to avoid degeneracy. Nevertheless, application to data sets involving longer time horizons remains of interest, as does a comparison with SMC schemes that use resample-move steps, such as SMC^2 (Chopin et al., 2013).

Supplementary Material

Supplementary material for Sequential Bayesian inference for stochastic epidemic models of cumulative incidence (DOI: 10.1214/25-BA1547SUPP; .pdf). Further synthetic data experiments: we assess the performance of the particle filter across multiple synthetic data sets. LNA details for the Ebola application: an intuitive derivation of the linear noise approximation of cumulative incidence under the assumption of an SEIR transmission model, followed by brief details of the inference scheme.

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